

Boundary Collision Dynamics: object-fit: contain

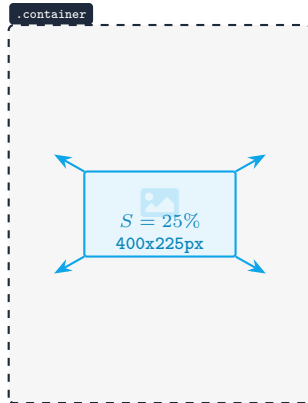
Real-time visualization of the interpolation engine explicitly halting upon edge contact

INTERPOLATION TIMELINE ($t_0 \rightarrow t_{collision}$)

Source Raster: 1600x900px (Aspect 16:9)

▶ $t = 0.25$ (Early Interpolation)

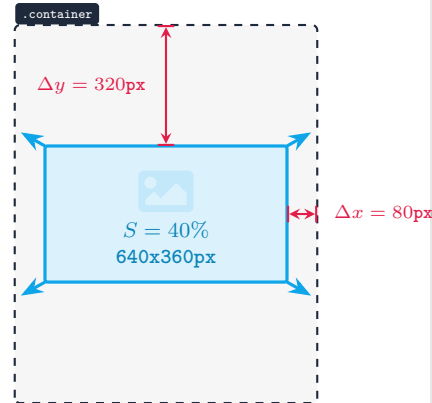
Layout Bounding Box ($800 \times 1000\text{px}$)



Status: Expanding freely.
Both X and Y dimensions are uniformly interpolated outward, processing pixels independently of container bounds.

▶▶ $t = 0.40$ (Mid Interpolation)

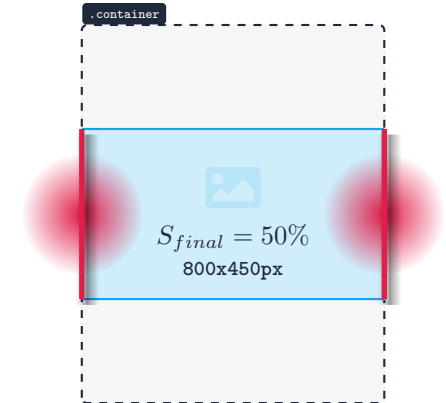
Layout Bounding Box ($800 \times 1000\text{px}$)



Status: Approaching boundaries.
The hardware acceleration calculates the remaining pixel delta to the Y-axis edge (Δy) and X-axis edge (Δx).

■ $t = 0.50$ ($t_{collision}$: Hard Halt)

Layout Bounding Box ($800 \times 1000\text{px}$)



Status: Edge boundary touched!
The X-axis collides with the container. The engine fires an immediate hard halt, preventing Y from filling the space, birthing the letterbox void.

⚠ THE MATHEMATICAL DERIVATIVE OF HALT

$$\frac{d}{dt}\text{Scale}(t) = 0 \quad \text{if} \quad W_{img}(t) = W_{cnt} \vee H_{img}(t) = H_{cnt}$$

The engine evaluates a strict logical OR operator. The exact processor cycle that *either* dimensional boundary is mathematically matched, scaling velocity drops to zero.